Sequential Causal Training of Windowed PINNs for the Lorenz ODE System

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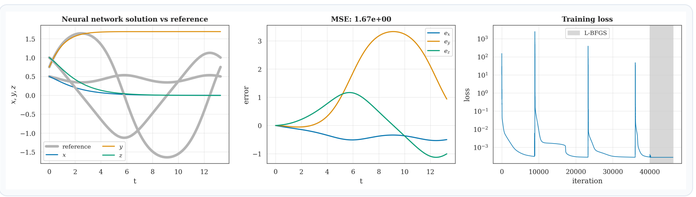
Final Year Design Project Defense

Code & saved run

The Long-Horizon Problem

Training one network per time window with causal weights cuts the combined RMSE from 2.237 to 6.97×10^{-5}, using only the ODE residual.

- A full-interval PINN couples all times, yet later states depend on earlier ones.
- Errors pile up and cannot be undone. Our saved single-network run stalls at RMSE 2.237.
- Builds on causal training (Wang et al., 2024).



One network, full interval. RMSE stalls at 2.237.

Lorenz-1960 IVP

GOVERNING EQUATIONS

$$\dot{x} = -0.1\,yz, \qquad \dot{y} = 1.6\,xz, \qquad \dot{z} = -0.75\,xy$$

Initial state u_0	(0.5, 0.75, 1.0)
Coefficients ($k=2,\ l=1$)	(−0.10, 1.60, −0.75)
Horizon	[0, 13.26446] · nearly closed orbit
Reference	DOP853 · rtol 10^{-10} · atol 10^{-12}
Evaluation	13,265 uniform times
Role of reference	monitoring only, never in the loss

- The reference ends almost where it starts: a **nearly closed orbit**, not an exact period.

Windowed PINN

- Each window k gets its own 4×60 tanh network. Windows share boundaries, not parameters.
- A hard trial solution fixes each initial state **by construction**, so there is no IC penalty to weight.
- Each window starts from the previous predicted endpoint and copies its weights as a warm start.

HARD TRIAL SOLUTION: EXACT IC

$$\hat{u}_k(t) = u_{0,k} + (t - t_k)\,N_k(t;\theta_k)$$

WINDOW HANDOFF + WARM START

$$u_{0,k+1} = \hat{u}_k(t_{k+1}), \qquad \theta_{k+1} \leftarrow \theta_k$$

Run Configuration

Interval / windows	[0, 13.26446] / 27 equal windows
Parameters	11,283 per window · 304,641 total
Collocation	1,536 uniform per window · 41,446 unique points
Optimizer	Adam 10^{-3} rate, ×0.9 every 1,000 steps
Causal stages	$\varepsilon = 0.01,\ 0.1,\ 1,\ 10$ · cap 4,000 steps
L-BFGS	≤ 1,000 closures per window
Precision / seed	float64 / seed 0

One Network vs 27 Windows

Measure	One network	27 windows
Trainable parameters	11,283	**304,641**
Training points	39,793 LHS	**41,446 uniform**
Combined state RMSE	2.237	**6.97×10^{-5}**
Max state-error norm	3.346	**3.09×10^{-4}**
Collocation residual MSE	2.80×10^{-4}	**1.87×10^{-7}**
Recorded wall time (s)	4,529	**881**

The two runs differ in several options at once, so no single factor is isolated.

34,765 Adam updates · ≤ 1,000 L-BFGS closures · 2 of 108 stages capped

Training Pipeline

1 **Problem setup**
$\dot{x} = -0.1\,yz, \quad \dot{y} = 1.6\,xz, \quad \dot{z} = -0.75\,xy, \qquad u_0 = (0.5,\ 0.75,\ 1.0)$
$t \in [0,\ 13.26446]$ · 27 windows · 1,536 points/window · float64

REPEAT FOR EACH WINDOW k = 1 … 27

2 **PINN for window k**
Input t · Hidden 4×60 tanh · Output $N_k(t) \in \mathbb{R}^3$ · Hard IC $\hat{u}_k = u_{0,k} + (t - t_k)\,N_k(t)$

3 **ODE residual**
$r_k = d\hat{u}_k/dt - f(\hat{u}_k)$, derivatives by autodiff

4 **Causal residual loss**
$\mathcal{L}_i = \mathrm{mean}(r_i^2), \quad w_i = \exp\left(-\varepsilon \sum_{j<i} \mathcal{L}_j\right), \quad \mathcal{L} = \mathrm{mean}(w_i \mathcal{L}_i)$
$\varepsilon:\ 0.01 \to 0.1 \to 1 \to 10$

5 **Adam optimization**
learning rate × 0.9 every 1,000 steps

Stage complete? $\min w > 0.99$ or 4,000 steps — NO → 2 · YES → All ε values used?

All ε values used? — YES → 6 · NO → Next ε (keep weights) → next ε

6 **L-BFGS**
≤ 1,000 iterations
unweighted residual MSE

7 **Save window**
checkpoint · history · prediction

Last window? $k = 27$ — NO → Next window: $u_{0,k+1} = \hat{u}_k(t_{k+1})$, copy weights, $k \leftarrow k+1$ → next window

YES →

8 **Reference evaluation**
DOP853 · 13,265 points · rtol 10^{-10} · atol 10^{-12}
used only for evaluation, never in training

9 **Evaluation outputs**
MAE · RMSE · max error · residual MSE · invariant drift · training cost

Per-window training loop. DOP853 is used only for evaluation.

Causal Weighting

- Each point's weight depends on the accumulated loss at all earlier points, so the network fits early times before late ones.
- Weights are recomputed every epoch and held fixed (detached) during the Adam step.

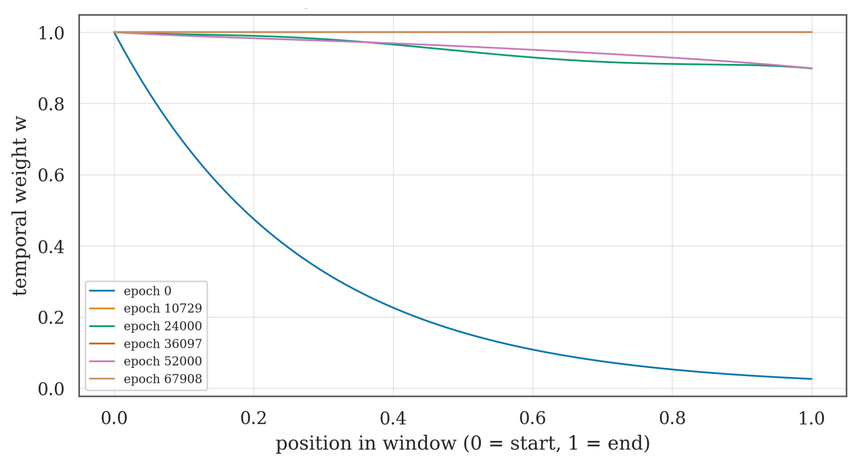
CAUSAL WEIGHT

$$w_{k,i} = \exp\left(-\varepsilon \sum_{j<i} \mathcal{L}_{k,j}\right)$$

CAUSAL LOSS

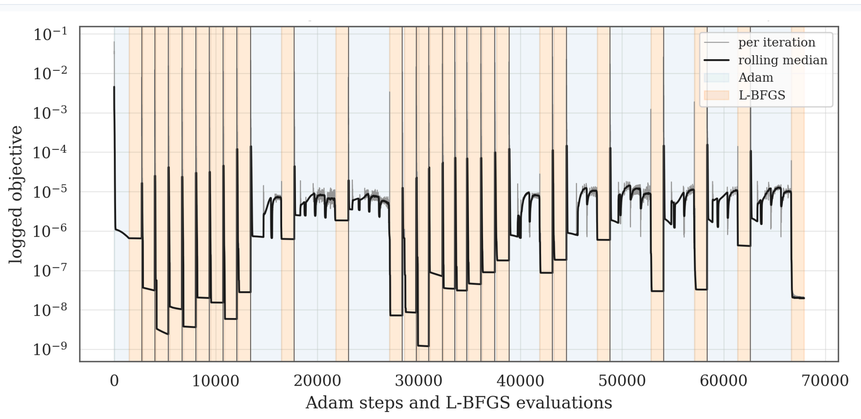
$$\mathcal{L}_{k,\varepsilon} = \frac{1}{n_k} \sum_{i=1}^{n_k} w_{k,i}\,\mathcal{L}_{k,i}$$

advance when $\min w > 0.99$ · then L-BFGS · unweighted residual mean



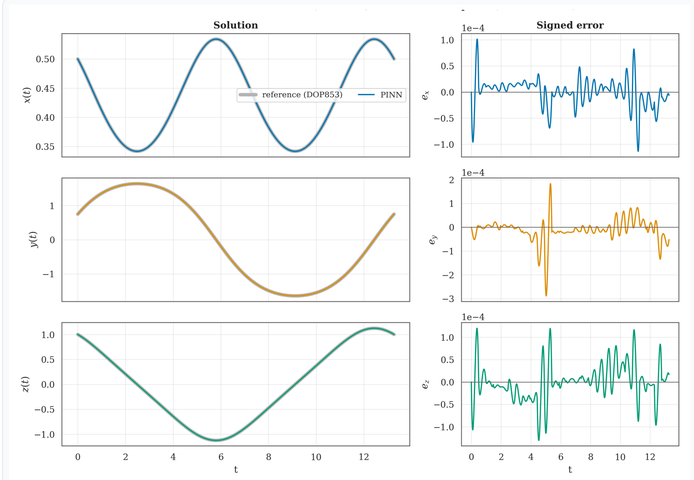
Causal weights by epoch. They flatten toward 1 as early losses shrink.

Optimizer Sequence



Adam, then L-BFGS, in each of the 27 windows. Loss falls by orders of magnitude.

Trajectory Accuracy



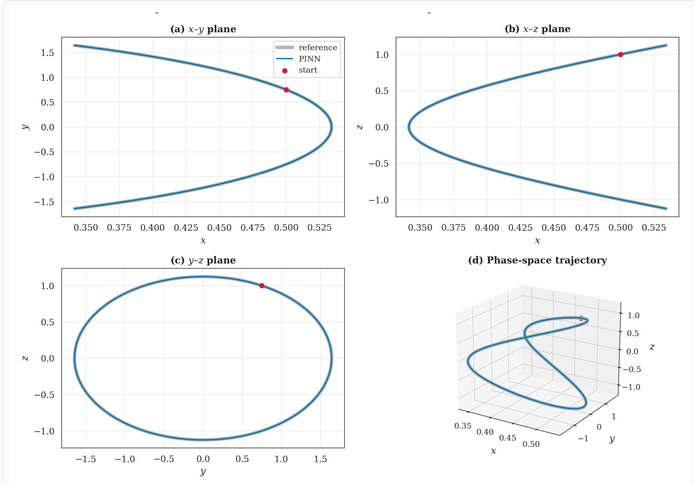
PINN vs DOP853. Signed errors (×10^{-4}) peak near $t \approx 5$.

State	MAE	RMSE	Max
x	1.89×10^{-5}	2.74×10^{-5}	1.13×10^{-4}
y	2.94×10^{-5}	4.96×10^{-5}	2.88×10^{-4}
z	2.78×10^{-5}	4.05×10^{-5}	1.30×10^{-4}
State norm	**5.12×10^{-5}**	**6.97×10^{-5}**	**3.09×10^{-4}**

Physics Checks

- Residual MSE: 1.874×10^{-7} on collocation points, 1.872×10^{-7} on evaluation points (gap ≈ 1.000).
- State error and ODE residual stay within 10^{-5}–10^{-3} along the whole orbit.
- Values are exactly continuous at all 26 inner interfaces, by construction.
- The DOP853 reference itself drifts by less than 4×10^{-10}.

Phase Portraits



PINN (blue) vs reference (grey). Red dot marks u_0.

Limitations & Future Work

- Each window after the first starts from a **predicted** state, so earlier error carries forward and derivatives can jump at interfaces.
- One seed, one initial state, one coefficient set. No windowed run without causal weights was saved.
- Next: one-change-at-a-time runs over several seeds, and boundary-value problems.

Lorenz (1960) Tellus · Raissi, Perdikaris & Karniadakis (2019) J. Comput. Phys. · Lagaris, Likas & Fotiadis (1998) IEEE Trans. Neural Netw. · Wang, Sankaran & Perdikaris (2024) Comput. Methods Appl. Mech. Eng. · Penwarden et al. (2023) J. Comput. Phys. · Matthews & Bihlo (2024) PinnDE · Virtanen et al. (2020) SciPy 1.0

Novelty

A reusable training algorithm for long-horizon ODE initial-value problems.

1. Split the horizon into windows with exact ICs.
2. Train each with staged causal weights + L-BFGS.
3. Hand endpoint and weights to the next window.

SOLVE YOUR OWN SYSTEM: SUPPLY

initial values u_0 · right-hand side $f(u)$ · time span $[0, T]$

github.com/ihmorol/lorenz1960-pinn